



Proteomics assessment of CSF and blood plasma of the PPMI cohort using Olink® Explore 1536 is a high-multiplex, high-throughput protein biomarker platform that uses Proximity Extension Assay (PEA) technology coupled to Next Generation Sequencing (NGS) using the Illumina® NovaSeq. Instrument

This document was prepared by retrieving documentation from Olink Proteomics website (olink.com) and borrowing information from the ppmi project 196

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PPMI Project ID: 214

Summary

Proteomics analysis was conducted in samples from cerebrospinal fluid (CSF) and blood plasma of 18 patients with Parkinson's Disease (PD) and 20 healthy controls (HC) from the PPMI cohort. The sample analysis here was intended to bridge analyses with another project on prodromal subjects. Analysis was conducted applying OLINK 1536 explore is covering 1536 analytes from inflammation, metabolic, cell regeneration, organ damage, neurodegeneration markers (such as neurofilament light chain) and other fields. Explore technology, which is based on two DNA tagged antibodies per analyte and amplification in a Proximity Extension Assay (PEA) for measuring protein levels ¹. The output of OLINK explore is the normalized protein expression (NPX) for 1536 targeted proteins. NPX displays a relative quantification, thus, the NPX values can only be compared for the same protein across the samples analyzed in one project and cannot be compared across different projects run at separate occasions without the application of bridging samples as a reference.

Associated Files

Data file: "biospecimen_results_holding_214_unblinded.csv"





Column headers glossary:

Variable	Description	Utility in Analysis
PATNO	Assigned patient ID	Patient ID in the ppmi database
COHORT	HC (Healthy Control); PD (Parkinson's Disease); Unk (OLINK Controls)	Group assignment of the sample
CLINICAL_EVENT	Visit for sample	Collection time point
TYPE	CSF or Plasma	Sample specimen source
SAMPLEID	Laboratory ID of the sample as chipped to OLINK	Sample ID, uniquely related to the PPMI patient ID
INDEX		
Olink_ID	Assay (protein being evaluated in a particular line); unique identifier on the vendor side	low, mostly for OLINK internal processing
UniProt	Six-digit alphanumeric code derived from the UniProt data base. Assays that lack a UniProt ID will have distinct syntax; Refer to https://www.uniprot.org/	Enabling rapid lookup of significant results obtained in an analysis. Naturally, it's difficult to have a strong command of 1400+ proteins; the Uniprot ID can help to identify the biology behind the OLINK results
Assay	Common name of the protein being assayed by OLINK explore	Main dataset descriptor together with the sampleID
MissingFreq	Data fraction (out of 1) beneath the limit of detection (see below) for a specific assay	Identification of low expressed proteins.
Panel	Allocation of the assay to a specific panel	Helps to find groups of proteins of interest
Panel_Lot_Nr	Panel lot number	
PlateI_D	Identifies the plate samples and assays were run	Enables checking the Normalization across plates





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Variable	Description	Utility in Analysis
QC_Warning	Quality control, Binary; identifies samples that passed OLINK QC	Olink recommends to exclude data with QC warning alerts from the data analysis
LOD	Limit of detection; defined as three standard deviations beneath the mean NPX value for a specific assay; in OLINK Explore, LOD is calculated for each plate individually	Olink recommends to exclude assays with a high proportion of samples below LOD from the analysis. Some combinations of LOD threshold with imputation methods for values below LOD may be required to run full intensity matrix in some statistical downstream analysis.
NPX	Normalized protein expression; calculated for each assay based on internal and external controls	Log 2 scale used in some statistical downstream analysis. Eventually after some LOD threshold or/and imputation of the values below LOD
NORMALIZATION	Method applied for data normalization	Here intensity normalization was implemented (See OLINK references)
ASSAY_WARNING	Warning mark, implicating possible problems with the assay	The analysis may be restricted to the Assays only without warning
RUNDATE	Date at which the results were handed to the principal investigator (May differ from the exact date the samples were run)	
Project_ID	Identification; Internal unique PPMI project ID	
PI_NAME	Name of the Principal investigator (PI)	
PI_INSTITUTION	Affiliation of the PI	The affiliation Brit Mollenhauer is Goettingen/Kassel (Germany); OLINK explore samples were run at the OLINK site in Upsala (Sweden)





Method

The raw data output of OLINK Explore is based on NGS (Next Generation Sequencing) counts, where every combination of an assay and sample is assigned to an integer value based on the number of detected DNA copies. These raw data counts get converted to NPX values; for details see: <https://www.olink.com/faq/how-is-the-npx-value-calculated-in-explore/>.

Extensive quality control measurements on each assay are run and reported by OLINK, allowing full visibility and control in regard of the technical performance at each step (see <https://www.olink.com/faq/how-is-quality-control-of-the-data-performed> and <https://www.olink.com/our-platform/our-pea-technology/data-generation-and-qc/>) for details.

The limit of detection (LOD) is defined as three standard deviations above the median normal protein expression (NPX) of the negative controls. Median per plate is set using all samples annotated as negative controls and a predefined standard deviation called fixSD gets applied. Detectability per assay and plate is defined as the percentage of samples displaying numbers above the LOD threshold. For details see <https://www.olink.com/faq/how-is-the-limit-of-detection-lod-estimated-and-handled/>.

Intensity normalization is the default for randomized studies on multiple sample plates. The median for a random sample selection displays a higher stability across different plates than the three plate controls (PC) on each plate. Intensity normalization sets the median level of all assays to the same value for all plates. For details see <https://www.olink.com/faq/data-normalization/> and <https://www.olink.com/content/uploads/2021/09/olink-data-normalization-white-paper-v2.0.pdf>.

References

1. Assarsson, E., *et al.* Homogenous 96-plex PEA immunoassay exhibiting high sensitivity, specificity, and excellent scalability. *PLoS One* 9, e95192 (2014).





Explore 384 Cardiometabolic (CARDIO) panel:

<https://www.olink.com/products/olink-explore/olink-explore-384-cardiometabolic/>

Explore 384 Inflammation (INF) panel: <https://www.olink.com/products/olink-explore/olink-explore-384-inflammation/>

Explore 384 Neurology (NEU) panel: <https://www.olink.com/products/olink-explore/olink-explore-384-neurology/>

Explore 384 Oncology (ONC) panel: <https://www.olink.com/products/olink-explore/olink-explore-384-oncology/>

The Olink Explore 1536 panel comprises of the panels, listed above: <https://www.olink.com/products/olink-explore/>

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